

Natural Language Input

"Simulate heat transfer
in a steel plate
at 500 K"



Domain Expert
or Researcher

PDE-Agents

Multi-Agent Orchestrator

LangGraph Supervisor Router

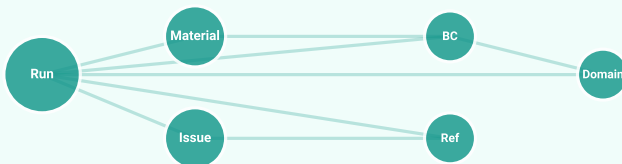
SIM
Simulation
Agent

ANA
Analytics
Agent

DB
Database
Agent

Local LLMs: Qwen3-Coder-Next | Llama 4 Scout | 2x RTX PRO 6000

GraphRAG Knowledge Base



Neo4j · 768-dim embeddings · HNSW vector search · warm-start injection

FEM Simulation Output

$$\rho c_p \frac{\partial u}{\partial t} - \nabla \cdot (k \nabla u) = f$$

POLENI · FEM-ICSx · PETSc KSP · θ-scheme · Gmsh geometries

$\mathcal{O}(h^2)$ verified

100%

Task Success
KG Smart (50 tasks)

97.8%

Production SR
1,369 real runs

1.00

MPF Score
Novel materials

2.9×

KG Advantage
vs. KG-free baseline

Three-Way KG Ablation

